

Archit Vishnoi

Portfolio: <https://v-archit.github.io> | Location: Rochester, NY

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Skills

Programming: C#, C++, C, SQL, Unreal Blueprints, OpenGL, DirectX, HTML, CSS, JavaScript, Linux.

Frameworks: Unity, Unreal, ASP.NET Core, MySQL, .NET MAUI, Blazor, RESTful APIs, GitHub, Perforce.

Concepts: Data Structures, Algorithms, System Design, Memory Management, Client-Server Architecture.

Practices: Performance Profiling, Code Reviews, Debugging, Agile/Scrum, Sprint Planning, Jira, Trello.

Experience

Newagez Technologies LLC, Missouri

Jan 2025 – Present

Software Engineer

- Designed scalable client-server network in C#, handling real-time state synchronization and fault tolerance.
- Built internal developer and automation tools, improving engineering productivity and reducing manual workflows.
- Integrated relational databases with backend services, optimizing SQL queries and improving data access latency.
- Bridged native C++ libraries with managed C# systems using P/Invoke, ensuring safe memory ownership, maintaining high-performance, and enabling seamless interoperability.
- Profiled and optimized CPU and memory usage in critical paths, increasing performance by 25%.

Archimic Studios LLC, Wisconsin

Feb 2024 – Dec 2024

Founder

- Architected and implemented large-scale C++ codebases with modular, testable components.
- Designed core systems, including input processing, simulation logic, AI state machines, and build pipelines.
- Wrote detailed technical documentation covering system architecture, data flow, and engineering trade-offs.

Vizara Technologies Pvt Ltd, Gurugram

May 2019 – Jun 2021

Computer Graphics Engineer

- Developed and shipped cross-platform applications including Digital Gandhi Akar Prakar and Chariot VR, emphasizing reliability and performance, achieving successful launches on the Google Play and Apple Store.
- Reduced memory consumption and runtime crashes by 20% through systematic profiling and optimization.
- Acted as Scrum Master, driving Agile adoption by leading sprint planning, code reviews, facilitating team collaboration, maintaining Git codebase and ensuring seamless communication using Jira.
- Implemented real-time virtual reality systems using C#, C++, and low-level graphics APIs of SteamVR.

Indian Institute of Technology, Delhi

Aug 2018 – Apr 2019

Junior Research Fellow

- Built interactive 3D systems with strict performance constraints and user experience and comfort requirements.
- Conducted applied research on long-session system stability, visualization, and optimization techniques.

Projects

Into The Babyverse – 3D Shooter Game – Unreal C++

2023

- Created a 6-degree-of-freedom spaceship system, custom hit physics, and object destruction using Chaos.
- Implemented physics-driven simulation, automatic Spline, and object lifecycle management system in C++.
- Designed data-oriented architectures with algorithmic optimizations to meet real-time performance constraints.
- Utilized Agile methodology in processes, including maintaining a comprehensive Game Design Document and conducting iterative evaluations through Sprints, ensuring consistent project alignment and timely delivery.

Hoodwink – 3D Wild West Action Game – Unreal C++

2022

- Developed AI system using state machine, Behavior trees, and graph-based decision logic for realistic actions.
- Optimized perception system scalability and interactivity by using custom BT tasks and Decorators.

The AI Story – 2D Narrative Game – Unity C#

2022

- Led development of programming, UI, logic creation, and dialogue systems with iterative sentence display.
- Streamlined team communication, boosting collaboration and reducing project timelines by 15%.

Tower Defense – 2D Realtime Strategy Game – Unity C#

2021

- Enhanced AI to prioritizing battles dynamically, increasing resource utilization and optimizing capacity deployment.
- Engineered a battle system ensuring 100% accuracy in combat status tracking and reducing processing latency.

Education

M.S. Game Design Development

Aug 2021 – Dec 2023

Rochester Institute of Technology, Rochester, NY

- Key coursework: Game Development Process, Game Design, Artificial Intelligence, Graphics Programming.

B.Tech. Computer Science Engineering

Aug 2014 – May 2018

Jaypee Institute of Information Technology, Noida